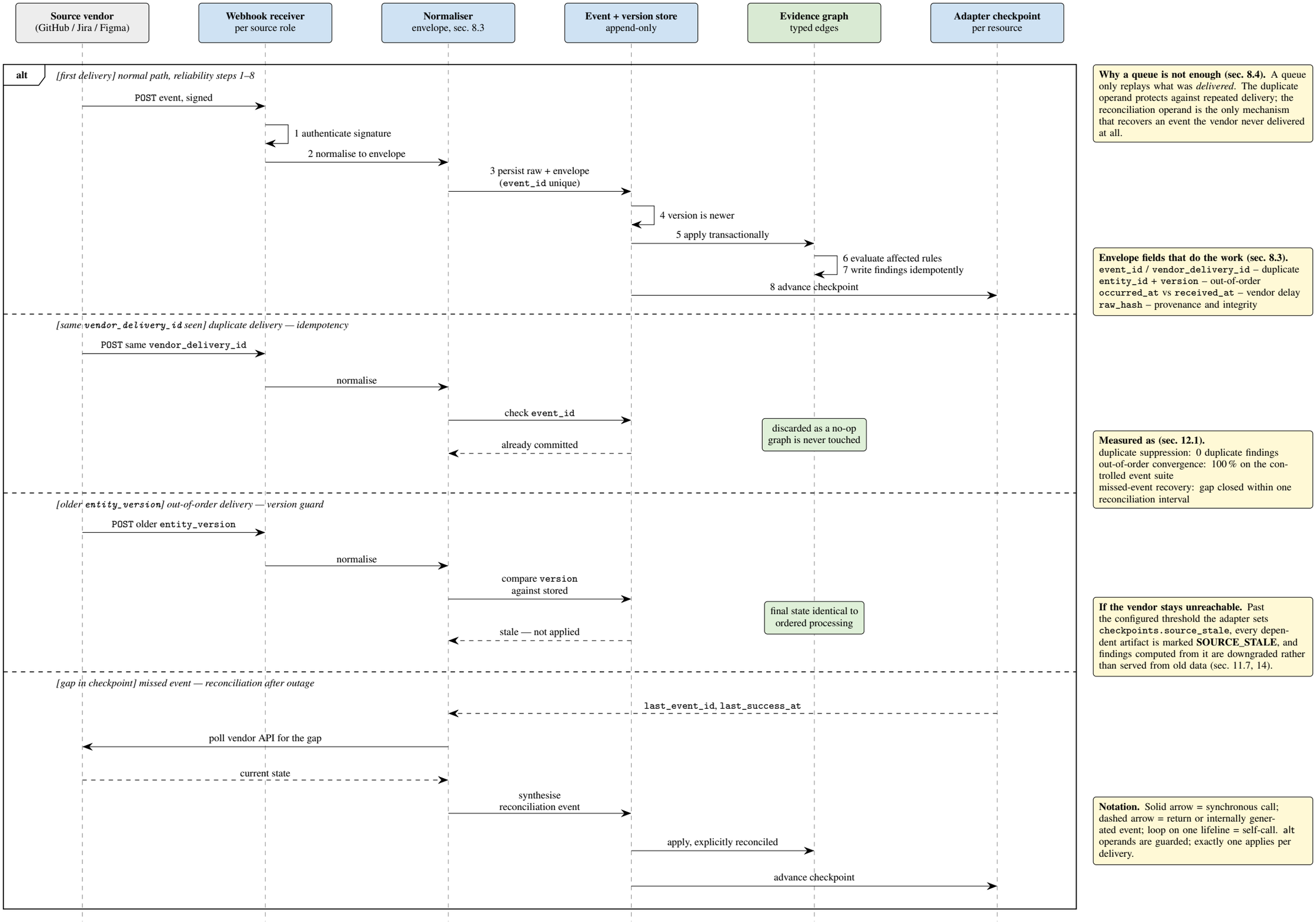


Why a queue is not enough (sec. 8.4). A queue only replays what was *delivered*. The duplicate operand protects against repeated delivery; the reconciliation operand is the only mechanism that recovers an event the vendor never delivered at all.

Envelope fields that do the work (sec. 8.3).
`event_id` / `vendor_delivery_id` – duplicate
`entity_id` + `version` – out-of-order
`occurred_at` vs `received_at` – vendor delay
`raw_hash` – provenance and integrity

Measured as (sec. 12.1).
duplicate suppression: 0 duplicate findings
out-of-order convergence: 100 % on the controlled event suite
missed-event recovery: gap closed within one reconciliation interval

If the vendor stays unreachable. Past the configured threshold the adapter sets `checkpoints.source_stale`, every dependent artifact is marked **SOURCE_STALE**, and findings computed from it are downgraded rather than served from old data (sec. 11.7, 14).

Notation. Solid arrow = synchronous call; dashed arrow = return or internally generated event; loop on one lifeline = self-call. `alt` operands are guarded; exactly one applies per delivery.